



Departamento de Engenharia Informática
Segurança em Tecnologias da Informação, MEI 2023/2024

Exam – Normal Season (1h45)

13 Jun 2024

1. Consider the email security solutions used on the Internet. Does using the DKIM mechanism to sign messages eliminate the need for PGP? Justify your answer
2. Consider that you want to guarantee confidentiality and integrity of data stored as records in a database. What algorithm(s) would you use for this purpose and, if applicable, in what mode? Justify your answer.
3. Consider that you want to use the IPSec protocol to guarantee the confidentiality and authentication for communications between two hosts located in two distinct networks, interconnected through a router which uses SNAT (Source NAT). Which header(s) would you use for this purpose? And in what mode (tunnel or transport)? Justify your answer.
4. Consider the Kerberos protocol. What is the role of the Authenticator in this protocol? Justify your answer.
5. Suppose Alice wants to send a message to Bob using the RSA algorithm. Bob already has a pair of keys, where his public key represented by (e,n) is $(3,55)$:
 - a) Encrypt the message represented by the number '20' using Bob's public key.
 - b) Is it possible to calculate Bob's private key? Justify your answer and, if yes, calculate it.

6. Consider security in wireless networks. Describe the role of the 802.1x protocol in wireless network security. Justify your answer.
7. Compare HMAC (Hash-based Message Authentication Code) algorithms with digital signature algorithms, with regard to ensuring integrity, authentication and non-repudiation. Justify your answer.
8. Consider using IPTables and Suricata for attack detection and prevention. Describe the role of inline mode in this context, in which the two technologies are used in conjunction. Justify your answer.